Конфигурационные файлы

docker-compose.yml

Name	Container ID	Image	Port(s)	CPU (%)	Memory usag...	Actions
airflow-docker	-	-	-	79.76%	2.63GB / 187.68(	
de-airflow-scheduler	1c000019327e	apache/airflow:		1.08%	275.2MB / 31.28	
de-airflow-web	1755067e8c56	apache/airflow:	8080:8080	76.88%	720.3MB / 31.28	
de-airflow-init	4bdd201c2e8d	apache/airflow:;		0%	0B / 0B	
de-nifi	921f4e3ff486	apache/nifi:1.23	8081:8080	1.08%	1.16GB / 31.28GI	
de-postgres	fc46c996e58c	postgres:14	5432:5432	0.46%	57.64MB / 31.28	
de-mongodb	a274436a3f66	mongo:7.0	27017:27017	0.26%	291.5MB / 31.28	
de-jupyter	34f3f37a20c2	continuumio/an	8888:8888	0%	164.1MB / 31.28	

init/mongodb/mongo-init.js — инициализация MongoDB (генерация данных)

init/de-postgres/init.sql — инициализация PostgreSQL

Инициализация баз данных

MongoDB (mongo-init.js)

service

Revina_service > service

Open MongoDB shell | Create collection | Refresh

Collection name	Properties	Storage size	Documents	Avg. document size	Indexes	Total index size
EventLogs	-	77.82 kB	1.5K	138.00 B	1	32.77 kB
ModerationQueue	-	53.25 kB	600	237.00 B	1	24.58 kB
SupportTickets	-	69.63 kB	465	568.00 B	1	20.48 kB
UserRecommendations	-	28.67 kB	300	161.00 B	1	20.48 kB
UserSessions	-	163.84 kB	1.5K	343.00 B	1	32.77 kB
Users	-	36.86 kB	300	194.00 B	1	20.48 kB

1. Users профили пользователей:

```
_id: ObjectId('69aa22c6caf7ef7c138563b1')
user_id : "user_1"
email : "user1@example.com"
name : "User 1"
registration_date : "2025-03-05T20:12:01.848Z"
last_active : "2025-10-17T17:19:22.557Z"
is_active : true
```

2. UserSessions, сессии пользователей:

```
_id: ObjectId('69ab6c66ad0f7bbcc88564e2')
session_id : "sess_006"
user_id : "user_200"
start_time : "2025-12-07T18:11:20.899Z"
end_time : "2025-12-07T19:49:20.899Z"
▾ pages_visited : Array (1)
    0: "/home"
device : "tablet"
▾ actions : Array (3)
    0: "remove_from_cart"
    1: "add_to_cart"
    2: "logout"
```

```
_id: ObjectId('69ab6c66ad0f7bbcc88564e3')
session_id : "sess_007"
user_id : "user_72"
start_time : "2025-11-12T20:59:56.507Z"
end_time : "2025-11-12T22:23:56.507Z"
▸ pages_visited : Array (7)
device : "tablet"
▸ actions : Array (2)
```

3. EventLogs, логи событий:

```
_id: ObjectId('69aa22c6caf7ef7c13856ab9')
event_id : "evt_0001"
timestamp : "2025-11-21T15:59:22.396Z"
event_type : "form_submit"
▾ details : Object
  ▸ /products/123 : Object
```

4. SupportTickets, обращения в поддержку:


```
_id: ObjectId('69aa22c6caf7ef7c1385709b')
ticket_id : "ticket_171"
user_id : "user_234"
status : "in_progress"
issue_type : "account_access"
▾ messages : Array (3)
  ▾ 0: Object
      sender : "user"
      message : "Товар не соответствует описанию"
      timestamp : "2025-11-15T02:07:15.271Z"
  ▾ 1: Object
      sender : "support"
      message : "Мы отправили вам инструкцию на email."
      timestamp : "2025-11-15T04:27:15.271Z"
  ▾ 2: Object
      sender : "user"
      message : "Спасибо, я понял. Проверю."
      timestamp : "2025-11-15T04:43:15.271Z"
created_at : "2025-11-15T02:07:15.271Z"
updated_at : "2025-11-15T04:27:15.271Z"
```


5. UserRecommendations, рекомендации пользователям:


```
_id: ObjectId('69aa22c6caf7ef7c13857266')
user_id : "user_1"
▾ recommended_products : Array (3)
    0: "prod_101"
    1: "prod_201"
    2: "prod_101"
last_updated : "2026-02-17T08:05:52.554Z"
```


6. ModerationQueue, очередь модерации отзывов:


```
_id: ObjectId('69aa22c6caf7ef7c13857394')
review_id : "rev_003"
user_id : "user_144"
product_id : "prod_401"
review_text : "Отличный товар!"
rating : 5
moderation_status : "pending"
▾ flags : Array (1)
    0: "contains_links"
submitted_at : "2026-02-19T06:50:02.369Z"
```

PostgreSQL (init.sql)

Создана БД service с таблицами:

Users — профили пользователей

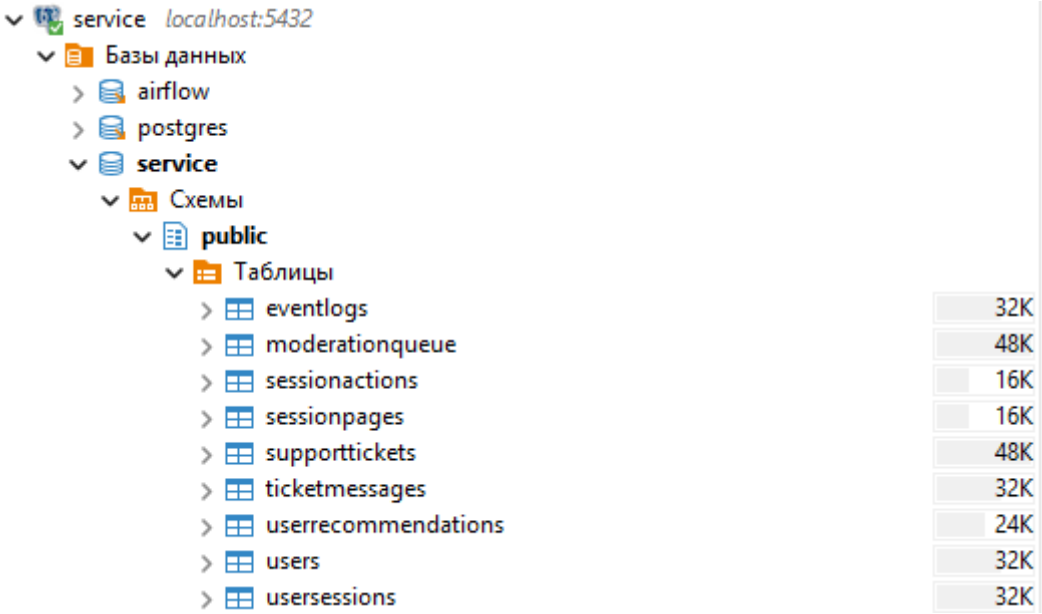
UserSessions, SessionPages, SessionActions — сессии пользователей

EventLogs — логи событий

SupportTickets, TicketMessages — обращения в поддержку

UserRecommendations — рекомендации пользователям

ModerationQueue — очередь модерации отзывов



Репликация данных

ETL-пайплайн для переноса данных из MongoDB в PostgreSQL (airflow/dags/etl_revina.py) содержит 6 задач для выгрузки, трансформации и загрузки коллекций: Users, UserSessions, EventLogs, SupportTickets, UserRecommendations, ModerationQueue.

На этапе выгрузки данные извлекаются из MongoDB с фильтрацией по дате для инкрементальной загрузки (при каждом запуске задачи выполняется запрос к целевой таблице для получения наибольшего значения created_at,

после чего из MongoDB извлекаются только те документы, у которых дата создания больше этого значения).

На этапе трансформации полученные документы преобразуются в pandas DataFrame, после чего выполняется очистка данных: приведение типов, удаление дубликатов, обработка пропущенных значений, нормализация вложенных структур (например, разворачивание массивов pages_visited, actions, messages в отдельные записи), преобразование словарей в JSON-строки и валидация некоторых данных.

Загрузка осуществляется в соответствующие таблицы PostgreSQL.

Демонстрация работы

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

user_id	email	name	registration_date	last_active	is_active	created_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

session_id	user_id	start_time	end_time	device	created_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

session_id	action_name	created_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

session_id	page_url	created_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

event_id	timestamp	event_type	{ } details	created_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

ticket_id	status	sender	message	timestamp	created_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

ticket_id	status	user_id	issue_type	created_at	updated_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

user_id	recommended_products	last_updated	created_at

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

Таблица

review_id	user_id	product_id	review_text	rating	moderation_status	flags	submitted_at	created_at

Airflow | DAGs | Cluster Activity | Datasets | Security | Browse | Admin | Docs | 21:39 UTC | AU

Do not use the **SequentialExecutor** in production. **Click here** for more information.

DAGs

All 3 | Active 0 | Paused 3 | Running 0 | Failed 0 | Filter DAGs by tag | Search DAGs | Auto-refresh

DAG	Owner	Runs	Schedule	Last Run	Next Run	Recent Tasks	Actions	Links
ETL_revina (etl, mongodb, postgres)	revina_uliana		0:30:00		2026-03-07, 21:00:00			
support_performance_revina (postgres, support_performance)	revina_uliana		0:30:00		2026-03-07, 21:00:00			
user_activity_revina (postgres, user_activity)	revina_uliana		0:30:00		2026-03-07, 21:00:00			

« ‹ 1 › » Showing 1-3 of 3 DAGs

Airflow | DAGs | Cluster Activity | Datasets | Security | Browse | Admin | Docs | 21:41 UTC | AU

DAG: ETL_revina | Schedule: 0:30:00 | Next Run ID: 2026-03-07, 21:30:00 UTC

07.03.2026 21:41:21 | All Run Types | All Run States | Clear Filters | Auto-refresh | 25

Press shift + / for Shortcuts

deferred | failed | queued | removed | restarting | running | scheduled | shutdown | skipped | success | up_for_reschedule | up_for_retry | upstream_failed | no_status

Duration 00:00:57 00:00:28 00:00:00 Mar 07, 21:39

etl_users
etl_user_sessions
etl_event_logs
etl_support_tickets
etl_user_recommendations
etl_moderation_queue

DAG ETL_revina

Details | Graph | Gantt | Code | Audit Log | Run Duration | Calendar

DAG Runs Summary	
Total Runs Displayed	5
Total success	5
First Run Start	2026-03-07, 21:39:26 UTC
Last Run Start	2026-03-07, 21:40:07 UTC
Max Run Duration	00:00:57
Mean Run Duration	00:00:46

Duration 00:00:57 00:00:28 00:00:00 Mar 07, 21:40

etl_users
etl_user_sessions
etl_event_logs
etl_support_tickets
etl_user_recommendations
etl_moderation_queue

Details | Graph | Gantt | Code | Audit Log | Logs | XCom | Task Duration

(by attempts)

1

All Levels | All File Sources

```
af28bc75faa2
*** Found local files:
***   * /opt/airflow/logs/dag_id=ETL_revina/run_id=scheduled__2026-03-07T21:00:00+00:00/task_id=etl_support_tickets/attempt=1.log
[2026-03-07, 21:30:27 UTC] {local_task_job_runner.py:120} ▶ Pre task execution logs
[2026-03-07, 21:30:27 UTC] {base.py:84} INFO - Using connection ID 'postgres_service' for task execution.
[2026-03-07, 21:30:27 UTC] {etl_revina.py:200} INFO - max data = 2025-01-01 00:00:00+00:00
[2026-03-07, 21:30:27 UTC] {base.py:84} INFO - Using connection ID 'mongo_default' for task execution.
[2026-03-07, 21:30:27 UTC] {etl_revina.py:243} INFO - etl_support_tickets done!
[2026-03-07, 21:30:27 UTC] {python.py:237} INFO - Done. Returned value was: True
[2026-03-07, 21:30:27 UTC] {taskinstance.py:441} ▶ Post task execution logs
```

Duration 00:00:57 00:00:28 00:00:00 Mar 07, 21:

etl_users
etl_user_sessions
etl_event_logs
etl_support_tickets
etl_user_recommendations
etl_moderation_queue

Details | Graph | Gantt | Code | Audit Log | Logs | XCom | Task Duration

(by attempts)

1

All Levels | All File Sources

```
af28bc75faa2
*** Found local files:
***   * /opt/airflow/logs/dag_id=ETL_revina/run_id=manual__2026-03-07T21:39:53.983594+00:00/task_id=etl_support_tickets/attempt=1.log
[2026-03-07, 21:40:18 UTC] {local_task_job_runner.py:120} ▶ Pre task execution logs
[2026-03-07, 21:40:18 UTC] {base.py:84} INFO - Using connection ID 'postgres_service' for task execution.
[2026-03-07, 21:40:18 UTC] {etl_revina.py:200} INFO - max data = 2026-03-05 07:46:07.407000
[2026-03-07, 21:40:18 UTC] {base.py:84} INFO - Using connection ID 'mongo_default' for task execution.
[2026-03-07, 21:40:18 UTC] {etl_revina.py:209} INFO - no data for etl
[2026-03-07, 21:40:18 UTC] {python.py:237} INFO - Done. Returned value was: True
[2026-03-07, 21:40:18 UTC] {taskinstance.py:441} ▶ Post task execution logs
```

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	user_id	email	name	registration_date	last_active	is_active	created_at
1	user_1	user1@example.c	User 1	2024-03-22 13:49:06.897	025-05-06 05:53:43.534	[v]	026-03-07 21:39:29.200
2	user_2	user2@example.c	User 2	2025-08-10 21:43:11.764	025-11-27 21:48:54.071	[v]	026-03-07 21:39:29.200
3	user_3	user3@example.c	User 3	2024-01-06 23:54:57.957	024-11-07 21:36:29.468	[v]	026-03-07 21:39:29.200
4	user_4	user4@example.c	User 4	2024-01-24 20:16:26.241	025-12-03 01:09:45.278	[v]	026-03-07 21:39:29.200
5	user_5	user5@example.c	User 5	2025-02-22 05:12:55.715	026-01-30 14:57:18.669	[v]	026-03-07 21:39:29.200
6	user_6	user6@example.c	User 6	2024-02-13 21:59:47.826	026-02-21 06:17:22.776	[v]	026-03-07 21:39:29.200
7	user_7	user7@example.c	User 7	2024-02-25 05:53:48.630	025-05-12 02:26:55.150	[v]	026-03-07 21:39:29.200
8	user_8	user8@example.c	User 8	2024-12-24 22:11:00.723	025-10-13 15:58:42.081	[v]	026-03-07 21:39:29.200
9	user_9	user9@example.c	User 9	2025-11-19 06:32:11.320	026-02-14 11:21:48.970	[v]	026-03-07 21:39:29.200
10	user_10	user10@example.c	User 10	2024-09-02 02:02:47.809	024-10-29 07:30:26.894	[ ]	026-03-07 21:39:29.200
11	user_11	user11@example.c	User 11	2025-05-26 17:26:25.661	025-07-02 16:24:51.477	[v]	026-03-07 21:39:29.200

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	session_id	user_id	start_time	end_time	device	created_at
1	sess_001	user_127	2026-01-16 13:02:58.957	2026-01-16 14:31:58.957	tablet	2026-03-07 21:39:31.914
2	sess_002	user_36	2026-01-25 19:06:16.784	2026-01-25 20:39:16.784	mobile	2026-03-07 21:39:31.914
3	sess_003	user_221	2025-11-23 10:46:11.002	2025-11-23 10:57:11.002	tablet	2026-03-07 21:39:31.914
4	sess_004	user_234	2025-11-07 14:33:21.206	2025-11-07 16:04:21.206	desktop	2026-03-07 21:39:31.914
5	sess_005	user_292	2025-11-27 00:25:38.171	2025-11-27 01:43:38.171	desktop	2026-03-07 21:39:31.914
6	sess_006	user_200	2025-12-07 18:11:20.899	2025-12-07 19:49:20.899	tablet	2026-03-07 21:39:31.914
7	sess_007	user_72	2025-11-12 20:59:56.507	2025-11-12 22:23:56.507	tablet	2026-03-07 21:39:31.914
8	sess_008	user_23	2026-01-02 23:47:56.005	2026-01-03 01:35:56.005	desktop	2026-03-07 21:39:31.914
9	sess_009	user_38	2026-01-21 10:38:53.276	2026-01-21 10:59:53.276	desktop	2026-03-07 21:39:31.914
10	sess_010	user_100	2026-02-23 19:52:38.924	2026-02-23 20:14:38.924	mobile	2026-03-07 21:39:31.914
11	sess_011	user_291	2026-01-05 00:44:01.079	2026-01-05 02:02:01.079	mobile	2026-03-07 21:39:31.914

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	session_id	action_name	created_at
1	sess_001	add_to_cart	2026-03-07 21:39:31.914
2	sess_001	logout	2026-03-07 21:39:31.914
3	sess_001	remove_from_cart	2026-03-07 21:39:31.914
4	sess_002	add_to_cart	2026-03-07 21:39:31.914
5	sess_002	logout	2026-03-07 21:39:31.914
6	sess_002	remove_from_cart	2026-03-07 21:39:31.914
7	sess_002	view_product	2026-03-07 21:39:31.914
8	sess_002	checkout	2026-03-07 21:39:31.914
9	sess_003	add_to_cart	2026-03-07 21:39:31.914
10	sess_004	logout	2026-03-07 21:39:31.914
11	sess_004	remove_from_cart	2026-03-07 21:39:31.914

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	session_id	page_url	created_at
1	sess_001	/cart	2026-03-07 21:39:31.914
2	sess_001	/about	2026-03-07 21:39:31.914
3	sess_001	/products/42	2026-03-07 21:39:31.914
4	sess_001	/contact	2026-03-07 21:39:31.914
5	sess_001	/checkout	2026-03-07 21:39:31.914
6	sess_002	/products/42	2026-03-07 21:39:31.914
7	sess_002	/products	2026-03-07 21:39:31.914
8	sess_002	/checkout	2026-03-07 21:39:31.914
9	sess_002	/cart	2026-03-07 21:39:31.914
10	sess_003	/products/123	2026-03-07 21:39:31.914
11	sess_003	/about	2026-03-07 21:39:31.914

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	event_id	timestamp	event_type	details	created_at
1	evt_0001	2026-02-07 07:36:41.446	click	{"/products": {}}	2026-03-07 21:39:36.401
2	evt_0002	2026-01-21 21:15:28.255	click	{"/home": {}}	2026-03-07 21:39:36.401
3	evt_0003	2026-01-12 15:11:22.121	click	{"/products/42": {}}	2026-03-07 21:39:36.401
4	evt_0004	2025-12-25 08:01:42.466	form_submit	{"/contact": {}}	2026-03-07 21:39:36.401
5	evt_0005	2026-01-07 00:51:34.401	page_view	{"/products/42": {}}	2026-03-07 21:39:36.401
6	evt_0006	2025-11-04 02:41:10.628	click	{"/about": {}}	2026-03-07 21:39:36.401
7	evt_0007	2025-11-07 03:04:18.750	form_submit	{"/about": {}}	2026-03-07 21:39:36.401
8	evt_0008	2025-12-02 11:27:29.200	scroll	{"/products": {}}	2026-03-07 21:39:36.401
9	evt_0009	2026-02-10 19:16:18.064	form_submit	{"/products/42": {}}	2026-03-07 21:39:36.401
10	evt_0010	2025-12-24 20:56:08.470	page_view	{"/products/123": {	2026-03-07 21:39:36.401
11	evt_0011	2026-03-04 16:45:12.483	scroll	{"/products": {}}	2026-03-07 21:39:36.401

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	ticket_id	status	sender	message	timestamp	created_at
1	ticket_032	open	user	Когда будет доставка? Уже прошла неделя	2026-01-15 22:54:24.748	2026-03-07 21:39:39.728
2	ticket_183	in_progress	user	Когда будет доставка? Уже прошла неделя	2025-12-09 07:24:09.070	2026-03-07 21:39:39.728
3	ticket_183	in_progress	support	Мы отправили вам инструкцию на email.	2025-12-09 09:08:09.070	2026-03-07 21:39:39.728
4	ticket_183	in_progress	user	Спасибо, я понял. Проверю.	2025-12-09 09:47:09.070	2026-03-07 21:39:39.728
5	ticket_063	open	user	Когда будет доставка? Уже прошла неделя	2025-12-13 08:59:46.990	2026-03-07 21:39:39.728
6	ticket_079	in_progress	user	Списалась двойная оплата	2026-02-14 03:04:31.370	2026-03-07 21:39:39.728
7	ticket_079	in_progress	support	Проблема решена. Проверьте, пожалуйста	2026-02-14 05:50:31.370	2026-03-07 21:39:39.728
8	ticket_079	in_progress	user	Спасибо, я понял. Проверю.	2026-02-14 06:14:31.370	2026-03-07 21:39:39.728
9	ticket_021	in_progress	user	Не могу оплатить заказ, пишет ошибку	2025-11-16 11:39:18.424	2026-03-07 21:39:39.728
10	ticket_021	in_progress	support	Здравствуйте! Мы уже проверяем вашу прс	2025-11-16 15:00:18.424	2026-03-07 21:39:39.728
11	ticket_079	open	user	Списалась двойная оплата	2026-02-14 03:04:31.370	2026-03-07 21:39:39.728

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	ticket_id	status	user_id	issue_type	created_at	updated_at
1	ticket_032	open	user_88	shipping	2026-01-15 22:54:24.748	2026-01-15 22:54:24.748
2	ticket_183	in_progress	user_282	refund	2025-12-09 07:24:09.070	2025-12-09 09:08:09.070
3	ticket_063	open	user_123	account_access	2025-12-13 08:59:46.990	2025-12-13 08:59:46.990
4	ticket_079	in_progress	user_97	shipping	2026-02-14 03:04:31.370	2026-02-14 05:50:31.370
5	ticket_021	in_progress	user_216	account_access	2025-11-16 11:39:18.424	2025-11-16 15:00:18.424
6	ticket_079	open	user_97	shipping	2026-02-14 03:04:31.370	2026-02-14 03:04:31.370
7	ticket_065	open	user_173	payment	2026-02-08 11:58:11.693	2026-02-08 11:58:11.693
8	ticket_025	open	user_88	payment	2026-01-20 11:24:50.975	2026-01-20 11:24:50.975
9	ticket_068	open	user_188	account_access	2025-12-16 11:14:48.509	2025-12-16 11:14:48.509
10	ticket_071	in_progress	user_176	shipping	2025-11-21 02:05:57.962	2025-11-21 05:16:57.962
11	ticket_016	resolved	user_121	shipping	2025-11-10 20:10:33.743	2025-11-11 02:20:33.743

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	user_id	recommended_products	last_updated	created_at
1	user_1	prod_202 [+2]	2026-03-01 00:51:49.238	2026-03-07 21:39:43.429
2	user_2	prod_103 [+2]	2026-02-20 13:43:15.867	2026-03-07 21:39:43.429
3	user_3	prod_102 [+2]	2026-02-13 05:56:48.163	2026-03-07 21:39:43.429
4	user_4	prod_201 [+2]	2026-03-05 13:38:16.952	2026-03-07 21:39:43.429
5	user_5	prod_302 [+2]	2026-02-14 22:05:18.051	2026-03-07 21:39:43.429
6	user_6	prod_301 [+2]	2026-02-20 09:24:08.700	2026-03-07 21:39:43.429
7	user_7	prod_103 [+2]	2026-02-12 00:17:58.219	2026-03-07 21:39:43.429
8	user_8	prod_401 [+2]	2026-02-22 18:13:22.882	2026-03-07 21:39:43.429
9	user_9	prod_202 [+2]	2026-02-04 15:35:47.787	2026-03-07 21:39:43.429
10	user_10	prod_401 [+2]	2026-03-04 14:44:26.666	2026-03-07 21:39:43.429
11	user_11	prod_301 [+2]	2026-02-08 05:44:56.520	2026-03-07 21:39:43.429

users 1 | usersessions 1 (2) | sessionactions 1 (3) | sessionpages 1 (4) | eventlogs 1 (5) | ticketmessages 1 (6) | supporttickets 1 (7) | userrecommendations 1 (8) | moderationqueue 1 (9)

select * from users; select * from usersessions; select * from s | Фильтрация данных не поддерживается

	review_id	user_id	product_id	review_text	rating	moderation_status	flags	submitted_at	created_at
1	rev_001	user_205	prod_302	Рекомендую	5	pending		2025-11-16 17:20:40.658	2026-03-07 21:39:46.156
2	rev_002	user_240	prod_101	Рекомендую	3	approved	contains_imag	2026-01-19 11:13:19.970	2026-03-07 21:39:46.156
3	rev_003	user_115	prod_102	Неплохо, но дорого	4	rejected		2026-02-07 17:05:18.313	2026-03-07 21:39:46.156
4	rev_004	user_155	prod_302	Отличный товар!	3	pending		2026-03-02 09:49:23.223	2026-03-07 21:39:46.156
5	rev_005	user_258	prod_103	Неплохо, но дорого	5	approved		2025-11-11 02:27:52.135	2026-03-07 21:39:46.156
6	rev_006	user_182	prod_102	Рекомендую	3	pending	spam	2026-02-11 13:06:29.526	2026-03-07 21:39:46.156
7	rev_007	user_253	prod_102	Не подошло	5	rejected		2025-12-26 08:27:41.224	2026-03-07 21:39:46.156
8	rev_008	user_76	prod_201	Неплохо, но дорого	2	rejected		2026-01-19 13:34:15.842	2026-03-07 21:39:46.156
9	rev_009	user_200	prod_401	Так себе	3	rejected	contains_links	2026-01-26 04:47:56.347	2026-03-07 21:39:46.156
10	rev_010	user_72	prod_202	Отличный товар!	3	rejected		2025-12-01 12:05:20.742	2026-03-07 21:39:46.156
11	rev_011	user_270	prod_202	Не подошло	4	rejected		2025-11-17 00:44:49.627	2026-03-07 21:39:46.156

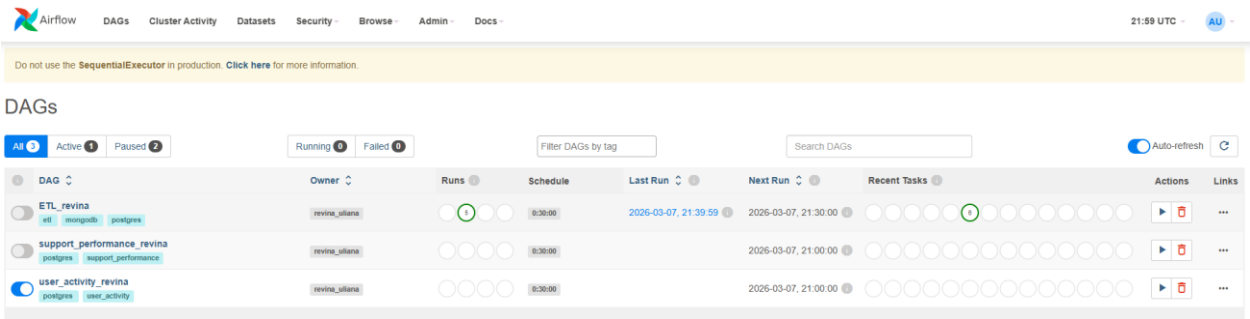
Аналитические витрины

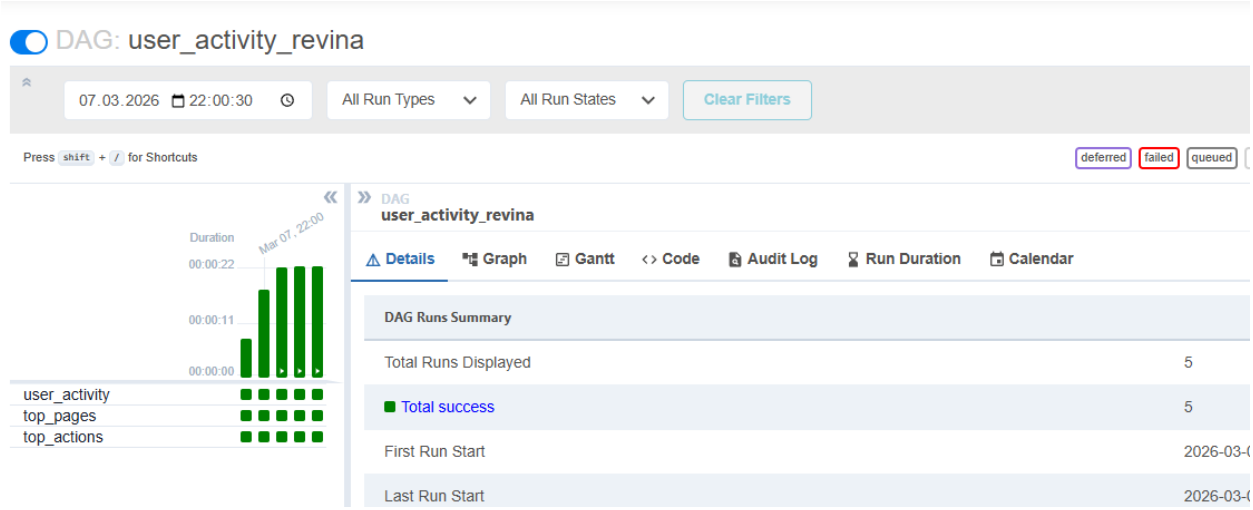
Витрина 1. Активность пользователей — поведенческий анализ (airflow/dags/user_activity.py)

Эта витрина должна определять, как пользователи взаимодействуют с сервисом — количество сессий, время на сайте, посещённые страницы и виды действий.

Цели:

- анализ активности пользователей по времени
- определение наиболее популярных страниц и действий





useractivity 1 | toppages 1 (2) | topactions 1 (3)

select * from UserActivity; select * from TopPages; select * frc | Фильтрация данных не поддерживается

	user_id	sessions_count	total_duration_minutes	pages_visited	actions_performed	last_active
1	user_291	13	8 904	/about [+7]	add_to_cart [+5]	2026-02-11 23:04:42.228
2	user_283	12	4 374	/about [+7]	add_to_cart [+5]	2026-02-21 12:15:31.302
3	user_280	12	6 009	/about [+7]	add_to_cart [+5]	2026-02-22 10:06:28.008
4	user_189	11	5 953	/about [+7]	add_to_cart [+4]	2026-02-22 22:50:46.744
5	user_73	11	8 716	/about [+7]	add_to_cart [+5]	2026-03-04 22:29:04.295
6	user_5	10	5 903	/about [+7]	add_to_cart [+5]	2026-02-24 02:19:05.324
7	user_16	10	5 598	/about [+7]	add_to_cart [+5]	2026-03-04 06:06:01.321
8	user_142	10	4 429	/about [+7]	add_to_cart [+5]	2026-02-22 07:03:06.824
9	user_221	10	4 392	/about [+7]	add_to_cart [+5]	2026-02-06 16:18:06.284
10	user_43	10	6 862	/about [+7]	add_to_cart [+5]	2026-02-23 13:43:36.368
11	user_71	10	5 219	/about [+7]	add_to_cart [+5]	2026-02-01 01:54:30.341
12	user_82	10	6 335	/about [+7]	add_to_cart [+5]	2026-03-02 01:16:55.235
13	user_286	10	5 564	/about [+7]	add_to_cart [+5]	2026-02-06 11:08:57.761
14	user_139	10	6 943	/about [+7]	add_to_cart [+5]	2026-02-18 12:12:16.077
15	user_14	10	7 519	/about [+7]	add_to_cart [+5]	2026-03-05 07:48:43.485
16	user_12	9	4 753	/about [+7]	add_to_cart [+5]	2026-02-26 01:10:10.081
17	user_143	9	6 620	/about [+7]	add_to_cart [+5]	2026-02-28 17:50:04.878
18	user_257	9	3 442	/about [+7]	add_to_cart [+5]	2026-03-02 01:29:58.859
19	user_273	9	6 235	/about [+7]	add_to_cart [+5]	2026-02-07 03:58:55.937

useractivity 1 | toppages 1 (2) | topactions 1 (3)

select * from UserActivity; select * from TopPages; select * frc

	page_url	views
1	/cart	653
2	/about	646
3	/home	640
4	/checkout	635
5	/contact	631

useractivity 1 | toppages 1 (2) | topactions 1 (3)

select * from UserActivity; select * from TopPages; select * frc

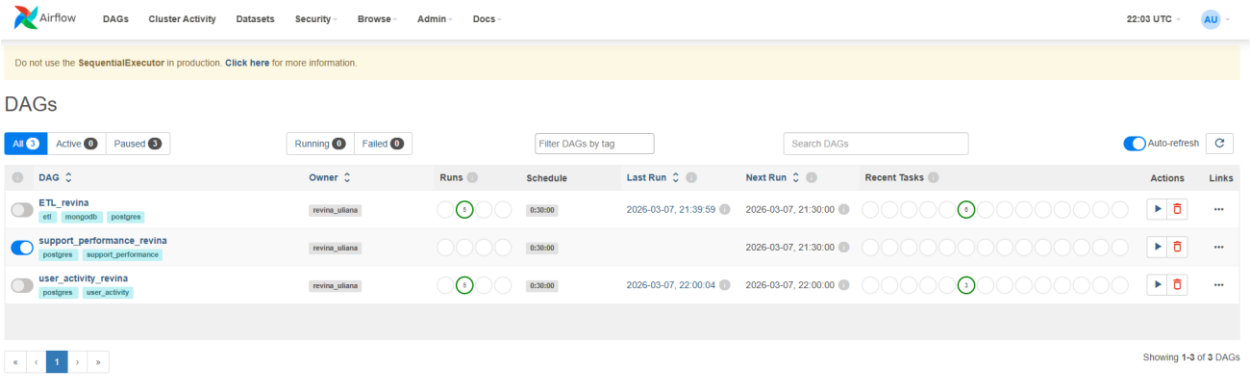
	action_name	actions_count
1	view_product	704
2	add_to_cart	703
3	logout	693
4	checkout	672
5	remove_from_cart	672

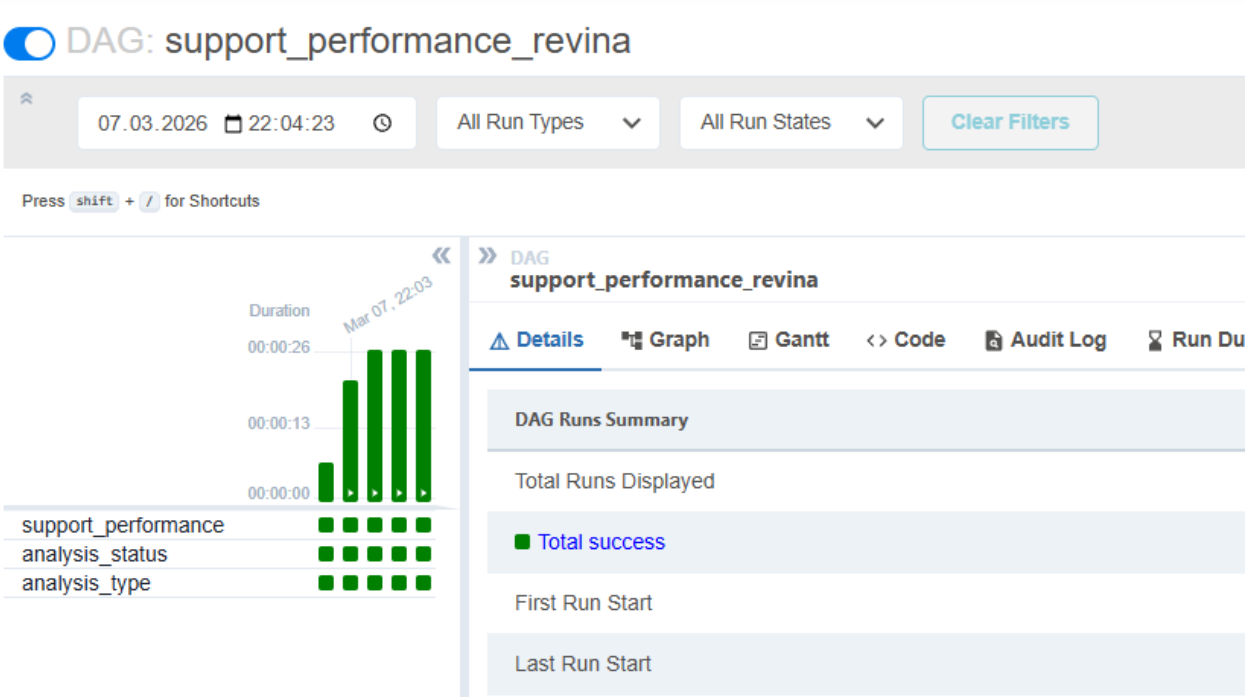
Витрина 2. Эффективность работы поддержки — статистика по обращениям (airflow/dags/support_performance.py)

Эта витрина должна отслеживать статус обращений, тип проблем и их актуальность.

Цели:

- подсчёт количества тикетов по статусам и типам
- время решения (от создания до обновления)
- анализ текущих открытых тикетов





supportperformance 1 | analysisstatus 1 (2) | analysistype 1 (3)

select * from SupportPerformance; select * from AnalysisStat | Фильтрация данных не поддерживается

Таблица | Текст

	ticket_id	user_id	issue_type	status	created_at	updated_at	lead_time
1	ticket_001	user_96	technical_issue	in_progress	2026-02-26 19:13:51.316	2026-02-26 22:35:51.316	13 130
2	ticket_002	user_59	shipping	open	2026-02-27 10:37:44.988	2026-02-27 10:37:44.988	12 206
3	ticket_003	user_148	technical_issue	open	2025-11-04 01:02:45.769	2025-11-04 01:02:45.769	178 381
4	ticket_004	user_35	technical_issue	resolved	2025-11-01 23:17:37.358	2025-11-02 05:26:37.358	369
5	ticket_005	user_161	payment	open	2026-01-27 21:20:00.233	2026-01-27 21:20:00.233	56 204
6	ticket_006	user_165	payment	resolved	2025-11-22 08:44:07.810	2025-11-22 16:04:07.810	440
7	ticket_007	user_23	technical_issue	open	2026-02-01 05:56:28.993	2026-02-01 05:56:28.993	49 928
8	ticket_008	user_260	account_access	open	2026-02-26 08:05:20.340	2026-02-26 08:05:20.340	13 799
9	ticket_009	user_58	account_access	resolved	2026-02-24 21:11:09.412	2026-02-25 04:57:09.412	466
10	ticket_010	user_177	refund	in_progress	2025-12-20 06:20:58.369	2025-12-20 08:28:58.369	111 823
11	ticket_011	user_277	shipping	open	2025-11-28 19:47:29.373	2025-11-28 19:47:29.373	142 697
12	ticket_012	user_146	shipping	resolved	2025-11-22 02:52:41.674	2025-11-22 08:43:41.674	351
13	ticket_013	user_154	account_access	resolved	2025-12-19 18:19:45.691	2025-12-20 01:27:45.691	428
14	ticket_014	user_216	payment	in_progress	2025-11-14 08:36:47.638	2025-11-14 10:46:47.638	163 527
15	ticket_015	user_180	shipping	open	2025-11-12 13:46:22.930	2025-11-12 13:46:22.930	166 098
16	ticket_016	user_121	shipping	resolved	2025-11-10 20:10:33.743	2025-11-11 02:20:33.743	370
17	ticket_017	user_191	payment	closed	2026-02-23 11:08:41.188	2026-02-23 20:54:41.188	586
18	ticket_018	user_17	shipping	open	2025-12-12 04:52:31.212	2025-12-12 04:52:31.212	123 432

supportperformance 1 | analysisstatus 1 (2) | analysistype 1 (3)

select * from SupportPerformance; select * from AnalysisStat | Фильтрация дан

Таблица | Текст

	status	tickets_count
1	open	60
2	closed	24
3	resolved	54
4	in_progress	62

supportperformance 1 | analysisstatus 1 (2) | analysistype 1 (3)

select * from SupportPerformance; select * from AnalysisStat | Фильтрация дан

Таблица | Текст

	issue_type	type_count
1	shipping	41
2	account_access	42
3	refund	46
4	payment	40
5	technical_issue	31